

Population vs. Sample

Population and **Sample** are two of the most fundamental concepts in statistics and data science. Almost every statistical analysis and machine learning project starts by understanding these concepts.

What is Population?

A **Population** is the **entire collection of individuals, objects, events, or observations** that we want to study.

It contains **all possible data** related to a particular problem.

DEFINITION

Population is the complete set of data or individuals about which we want to draw conclusions.

CHARACTERISTICS OF POPULATION

- Includes every member of the group.
 - Usually very large.
 - Can be finite or infinite.
 - Represented by **N** (population size).
 - Population parameters are usually unknown.
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EXAMPLES OF POPULATION

Study	Population
Student marks	All students in a university
Election survey	All eligible voters in India

Hospital study	All patients in a hospital
Online shopping	All customers of Amazon
Company employees	Every employee in the company

EXAMPLE 1

A university has **20,000 students**.

If you want to know the average height of students:

Population = All 20,000 students

EXAMPLE 2

An e-commerce company wants to know customer satisfaction.

Population = Every customer who purchased from the website

What is Sample?

A **Sample** is a **small subset of the population** selected for analysis.

Instead of collecting data from everyone, we collect data from a representative group.

DEFINITION

A Sample is a smaller group selected from the population to represent the whole population.

CHARACTERISTICS OF SAMPLE

- Smaller than the population.
- Easier to collect.
- Saves time and money.

- Represented by **n** (sample size).
- Used to estimate population parameters.

EXAMPLES OF SAMPLE

Populatio n	Sample
20,000 students	500 students
All voters	2,000 voters
All employee s	150 employee s
All customers	1,000 customers

Why Do We Use a Sample?

Studying the entire population is often difficult because it may be:

- Very large
- Expensive
- Time-consuming
- Sometimes impossible

A well-chosen sample provides results that closely represent the population.

EXAMPLE

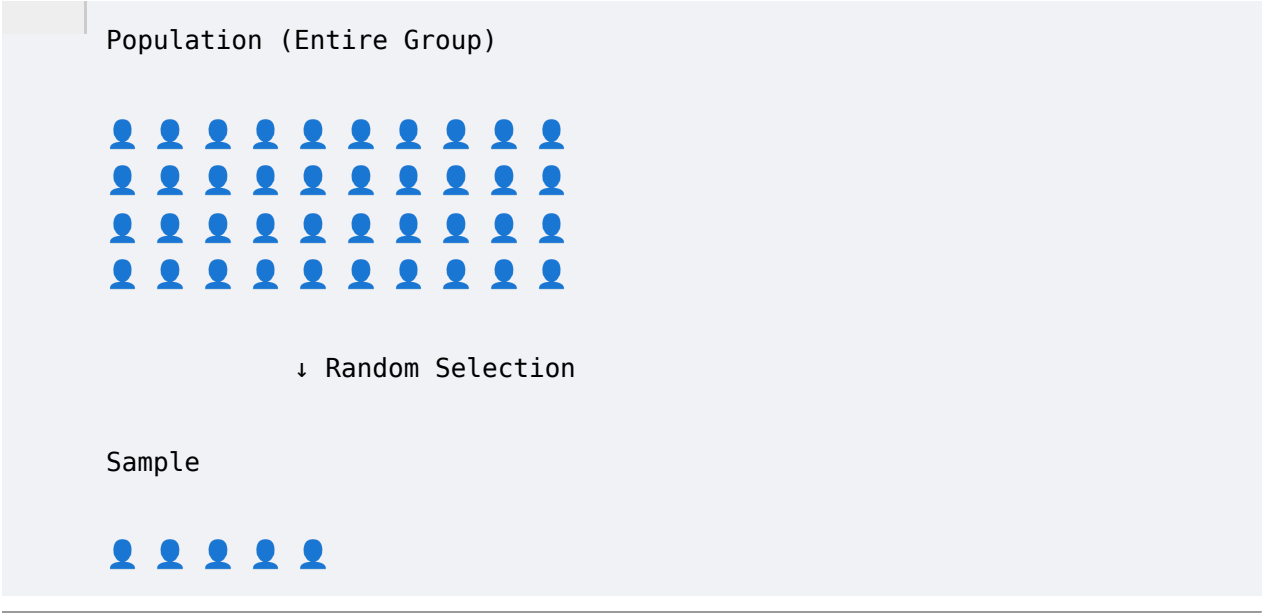
India has over **1.4 billion** people.

It is impossible to ask every person which political party they support.

Instead:

- Survey **2,000–5,000 people** (sample).
- Use the results to estimate the opinion of the entire population.

Population vs. Sample Diagram



Population Parameters vs. Sample Statistics

Populatio n	Sample
Parameter	Statistic
Fixed value	Estimated value
Usually unknown	Calculate d from data

COMMON SYMBOLS

Populatio n Paramete r	Symbol	Sample Statistic	Symbol

Population Mean	μ (mu)	Sample Mean	$\bar{x}$ (x-bar)
Population Standard Deviation	σ (sigma)	Sample Standard Deviation	s
Population Variance	σ^2	Sample Variance	s²
Population Size	N	Sample Size	n
Population Proportion	P	Sample Proportion	$\hat{p}$ (p-hat)

Real-World Examples

EXAMPLE 1: COLLEGE

Problem: Find the average marks of students.

- **Population:** All 10,000 students in the college.
- **Sample:** 500 randomly selected students.

EXAMPLE 2: MANUFACTURING

A factory produces **50,000 bulbs** every day.

Testing every bulb would be costly and time-consuming.

- **Population:** All 50,000 bulbs.
- **Sample:** 300 bulbs selected for quality testing.

EXAMPLE 3: HEALTHCARE

A hospital wants to estimate the average blood pressure of patients.

- **Population:** All patients in the hospital.

- **Sample:** 200 randomly selected patients.
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EXAMPLE 4: DATA SCIENCE

A movie streaming platform wants to know whether users like a new recommendation system.

- **Population:** All users of the platform.
 - **Sample:** 5,000 randomly selected users whose behavior is analyzed.
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Types of Population

1. FINITE POPULATION

A population with a fixed number of members.

Examples:

- Students in a class
 - Employees in a company
 - Books in a library
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2. INFINITE POPULATION

A population with no practical limit.

Examples:

- Outcomes of rolling a die repeatedly
 - Daily website visitors over an unlimited future period
 - Continuous sensor readings
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Types of Samples

1. RANDOM SAMPLE

Every member has an equal chance of being selected.

Example: Randomly choosing 100 student IDs from a list.

2. STRATIFIED SAMPLE

The population is divided into groups (strata), and samples are taken from each group.

Example: Selecting students from each department proportionally.

3. SYSTEMATIC SAMPLE

Select every **k-th** item after a random starting point.

Example: Checking every 10th product on an assembly line.

4. CLUSTER SAMPLE

The population is divided into clusters, and one or more entire clusters are selected.

Example: Selecting a few schools and surveying every student in those schools.

Population vs. Sample Comparison

Feature	Populatio n	Sample
Meaning	Entire group	Subset of the group
Size	Large	Smaller
Symbol	N	n
Cost	High	Lower
Time	More	Less
Accuracy	Highest (if measured correctly)	Depends on how representative the sample is
Purpose	Complete information	Estimate population characteristics

Importance in Data Science

Population and sample concepts are essential because:

- Collecting data from everyone is often impractical.
 - Machine learning models are trained on **sample data**.
 - Insights from the sample are used to make predictions about the **population**.
 - Good sampling improves the accuracy and reliability of analyses and models.
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Summary

Population	Sample
Entire dataset or group	Small subset of the dataset
Represented by N	Represented by n
Described by parameters (μ , σ , σ^2)	Described by statistics ($\bar{x}$, s , s^2)
More expensive and time-consuming to study	Faster and more cost-effective
Used when complete data is available	Used to estimate characteristics of

	the population
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KEY TAKEAWAY

- **Population = Whole group you want to study.**
- **Sample = Smaller group selected from the population.**
- **In Data Science, we usually collect a representative sample and use statistical methods to make reliable conclusions about the entire population.**